Project Report Of Natural Language Processing

"Image to Text Dataset for Quantum Computing"

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Abstract

The automatic creation of image–text datasets for scientific schematic diagrams is a non-trivial task, as such diagrams differ fundamentally from natural images and are insufficiently handled by generic image-to-text models. This project investigates the feasibility of compiling a specialized dataset of quantum circuit diagrams from scientific publications with minimal manual effort. The objective is to construct a prototypical dataset consisting exclusively of quantum circuit images extracted from recent arxiv[1] papers in the quant-ph category, together with structured metadata describing their visual and textual context. Particular emphasis is placed on reproducibility, generalizability, and strict adherence to dataset construction constraints.

To address this task, an automated pipeline is developed that combines document layout analysis, geometric heuristics, image-based classification, optical character recognition, and textual relevance filtering. Candidate figures are identified directly from PDF documents, including vector-based diagrams, and filtered to isolate quantum circuit schematics from other graphical content. For each extracted image, contextual information such as figure identifiers, gate occurrences, associated quantum problems, and descriptive text passages is derived from the surrounding document structure. The approach builds upon established concepts from document intelligence and combined visual–text extraction while adapting them to the specific challenges posed by schematic quantum circuits. The results demonstrate that constructing a structured image–text dataset for quantum circuits from scientific literature is feasible, while also revealing limitations arising from diagram variability and semantic ambiguity in technical texts.

1 Introduction

Image-text datasets are essential for multimodal learning, yet comparable resources for scientific schematics remain limited. Quantum circuit diagrams used to represent quantum algorithms and experimental setups are especially challenging for generic image-to-text models because they are highly structured, often vector-based, and use domain-specific notation [7, 6]. Building specialized datasets is therefore important, but manual annotation is costly and does not scale.

This project investigates whether a structured dataset of quantum circuit images and textual metadata can be compiled automatically from scientific papers under strict reproducibility and minimal manual effort constraints. It develops a pipeline combining document layout analysis, computer vision, and NLP to extract circuit figures, filter non-circuit graphics, and derive metadata such as page/figure identifiers, gate information, and descriptive text passages. The work emphasizes robustness and transparency, and analyzes typical error modes (e.g., false positives, missed figures, and ambiguous descriptions) to evaluate feasibility for larger-scale collection.

2 Problem Statement and Key Objectives

Quantum computing papers rely heavily on circuit diagrams to communicate algorithms and procedures, yet such schematics are poorly covered in existing image–text datasets and are not well handled by generic captioning models trained on natural images. The core problem addressed in this project is the lack of scalable, reproducible methods to automatically construct structured image–text datasets of quantum circuit diagrams directly from scientific PDFs, given heterogeneous layouts, frequent vector-based figures, and the weak or ambiguous linkage between figures and surrounding technical text.

The key objectives of this project are: (i) evaluate the feasibility of automatically extracting circuit images and associated textual metadata from scientific publications, (ii) design a general and reproducible pipeline under strict data-source and processing-order constraints, and (iii) analyze typical error cases and quality limitations of automated extraction. Overall, the project is a feasibility study emphasizing robustness and dataset quality rather than exhaustive semantic interpretation.

3 Methodology

The methodology follows a multi-stage pipeline that processes arXiv papers one-by-one in the prescribed order. Each PDF is parsed page-by-page to extract figure candidates, including vector-based diagrams, which are then refined through layout aware grouping and geometric filtering before classification to isolate quantum circuit schematics. For each retained circuit image, the pipeline extracts structured metadata page/figure identifiers, detected gates, a problem label, and a caption/context description with traceable text positions—and saves the image together with its JSON entry. Processing is deterministic and stops once the target number of circuit images is reached, ensuring reproducible results across papers and layouts.

`Input: arXiv papers list (paper_list_4.txt)`

↓

1. PDF Acquisition & Setup (`data_acq_px*.py`)

Download PDFs (arXiv API), create folders, resume-safe processing

↓

2. PDF Parsing & Page Iteration (`fitz / PyMuPDF`)

Iterate pages; extract text blocks + vector drawings; render page for crops

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3. Figure Candidate Region Extraction (`circuit_extr_px.py`)

Cluster/merge vector elements into candidate boxes; crop candidate images

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4. Multi-stage Filtering (Rejection Gates) (`text_layout_utils.py`)

Reject tiny/noisy regions; remove prose/bibliography-like blocks using geometry + text heuristics

↓

5. Circuit vs Non-circuit Classification (`circuit_classifier.py`)

CNN scoring + conservative thresholds; discard non-circuits and low-confidence candidates

↓

6. Metadata Extraction (`quantum_extr_px.py, text_layout_utils.py`)

Extract page/figure number, caption; detect gates + problem label; select nearby description + text positions

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7. Save Dataset Artifacts (`JSON + PNG`)

Write circuit image to disk; append JSON entry; update per-paper image-count CSV

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Stop Condition

Terminate deterministically once 250 circuit images are saved

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`Output: Circuit PNG images (images_4) + metadata JSON (dataset_4.json) + counts CSV (paper_list_counts_4.csv)`

Figure 1: Dataset construction pipeline for extracting quantum circuit figures and metadata from arXiv PDFs.

3.1 PDF acquisition & figure candidate extraction

Scientific papers are retrieved from the arXiv[1] platform using the arXiv API based on a predefined and ordered paper list (see `load_paper_list()` and `download_pdf()` in `build_dataset.py`). The corresponding PDF files are downloaded and processed locally to ensure reproducibility and adherence to the prescribed processing order, and all subsequent stages operate exclusively on these downloaded documents.

Figure candidate extraction is performed directly on the structural representation of PDF documents to capture vector-based figures independently of their raster rendering[9][10]. Since quantum circuit diagrams are frequently encoded as vector drawings rather than embedded images, the pipeline first detects pages containing drawing primitives and then forms candidate regions from clustered drawing bounding boxes (see `extract_and_classify_clusters()` in `extract_circuit_images.py`, which uses `page.get_drawings()` and `page.cluster_drawings()`). Nearby drawing clusters are merged into larger candidate regions to prevent fragmented circuit components from being treated as separate figures (see `merge_nearby_rects()` in `extract_circuit_images.py`). These regions are treated as figure candidates without semantic interpretation, with the objective of maximizing coverage by retaining all potential figures for subsequent refinement.

3.2 Quantum Circuit Image Identification and Classification

This stage determines whether a figure candidate represents a valid quantum circuit diagram and should be retained in the dataset.

3.2.1 Spatial Grouping of Vector Drawing Primitives

Quantum circuit figures in scientific PDFs are often encoded as many separate vector drawing operations (wires, gate boxes, connectors) rather than a single embedded image. The pipeline therefore groups drawing primitives into page-local clusters directly from the PDF drawing structure (see the call to `page.cluster_drawings()` inside `extract_and_classify_clusters()` in `extract_circuit_images.py`). These clusters are then merged at the bounding-box level into coherent candidate figure regions by combining nearby or overlapping rectangles based on proximity and overlap (see `merge_nearby_rects()` in `extract_circuit_images.py`).

This design exploits the geometric structure of vector-based circuit diagrams and avoids rasterizing full pages, which is more sensitive to resolution and introduces noise from surrounding text. A limitation is occasional over-merging of nearby but unrelated elements, which is handled later by circuit-specific filtering and classification stages.

3.2.2 Page-Local Figure Region Formation

After spatial grouping, page-local clusters are merged into candidate figure regions represented as bounding boxes (see `merge_nearby_rects()` as used inside `extract_and_classify_clusters()` in `extract_circuit_images.py`). Processing is strictly page-by-page, so regions are formed independently per page, preventing accidental merges across page boundaries and keeping results deterministic and traceable (see the per-page loop in `extract_and_classify_clusters()` in `extract_circuit_images.py`).

Large or composite regions may still include nearby non-circuit content (e.g., captions), which is addressed later through crop refinement rather than at the grouping stage (see `trim_bbox_using_pdf_text()` in `extract_circuit_images.py`).

3.2.3 Rule-Based Circuit Filtering

After figure-region formation, many candidates still correspond to non-circuit content, most commonly dense text blocks (e.g., references, footnotes) or regions where captions/prose were merged into the same drawing cluster. To reduce these obvious false positives before applying machine learning, the pipeline uses a deterministic rule-based pre-filter that rejects clearly text-like regions using clipped PDF text signals (see `is_text_by_content()` in `extract_circuit_images.py`). An additional image-level heuristic rejects candidates that visually resemble paragraph text (dense stroke texture / text-like structure) rather than schematic graphics (see `is_text_by_geometry()` in `extract_circuit_images.py`). Candidate crops are also refined to reduce caption leakage by trimming overlapping "Figure ..." lines or nearby textual artifacts when detected in the PDF text layer (see `trim_bbox_using_pdf_text()` in `extract_circuit_images.py`). This pre-filter is chosen because its criteria are fully deterministic and require no additional training data, improving reproducibility and reducing computational cost. It also prevents the CNN from being dominated by trivial non-figure regions, while still deferring ambiguous graphics to the image-based classifier in the next stage.

3.2.4 Image-Based Circuit Classification Using a Fine-Tuned CNN

An image-based classification model is applied to candidates that remain after deterministic pre-filtering to distinguish quantum circuit diagrams from visually ambiguous non-circuit figures (see the filtering and inference flow in `extract_and_classify_clusters()` in `extract_circuit_images.py`, where candidates are rejected by `is_text_by_content()` / `is_text_by_geometry()` before model inference). The final candidate crop is rendered and classified using `predict_image()` in `circuit_classifier.py`, which returns the predicted label and confidence.

For classification, a ResNet18[5] convolutional neural network is used for binary prediction (circuit vs non-circuit). The model is fine-tuned on a balanced dataset curated from open sources: quantum circuit images collected from a Kaggle[3] dataset and non-circuit scientific figures collected from the DocFigure dataset[4], subsampled to approximately 500 images per class. This curated dataset is split into train/validation/test directories and loaded directly using PyTorch ImageFolder.

Transfer learning is used to adapt ImageNet features to circuit schematics. Training is performed in two stages: first training only the new classification head, and then fine-tuning the last ResNet block together with the head to improve domain adaptation while keeping computation light (see the Phase 1 / Phase 2 training procedure in main() in `train_circuit_classifier.py`). During dataset compilation, the trained model is evaluated on the images produced by the extraction pipeline after the rule-based filters, i.e., the candidate crops that survive content/geometry filtering and are then classified and thresholded by probability before being saved.

3.3 Textual and Metadata Extraction

3.3.1 Page and Figure Number Resolution

For each figure classified as a quantum circuit, the pipeline extracts page and figure identifiers directly from the PDF. Document page numbers are resolved using a footer-based approach: for each page, a fixed footer band is clipped and parsed using regex rules. If no reliable printed page number is detected, the pipeline assigns `page_index+1` to guarantee an integer page number for every saved record (see the `page_numberassignment` inside `extract_and_classify_clusters()` in `extract_circuit_images.py`).

Figure numbers and caption text are extracted with a caption-first, layout-aware search around the final figure crop. The function `detect_figure_number_and_caption()` scans text lines in a vertical window around the figure, searching below first (most common caption placement) and then above, extracts candidate figure labels using multi-pattern regex matching in `extract_figure_number_from_text()`, and selects the closest matching line by geometric distance; caption text is then derived from that line using `extract_caption_from_text()` (see these methods in `text_layout_utils.py`). Since the dataset specification requires integer-only figure identifiers, extracted labels (which may be strings such as 3a, 3.1, or a) are normalized deterministically when writing the JSON: numeric labels are preserved, numeric+suffix labels are encoded as `base*100+suffix_index` (e.g., 3a $\rightarrow$ 301), and non-resolvable purely alphabetic labels are mapped to a sentinel -1 (see `normalize_figure_number()`in `build_dataset.py`).

3.3.2 Quantum Gate Extraction from Images and Text

Quantum gate extraction follows a strict, image-aware priority order. The pipeline first runs OCR on the circuit image (see `extract_gates_from_image()` in `gate_problem_extractor.py`); the crop is grayscaled and binarized, OCR is executed with `--psm6`, and tokens are matched against a gate lexicon (see `MULTI_CHAR_GATES` and `SINGLE_CHAR_GATES` in `gate_problem_extractor.py`). To avoid variable–gate confusion, OCR single-letter gates are accepted only when uppercase (see the `token.isupper()` check in `extract_gates_from_image()`). If OCR yields at least one gate, it is treated as authoritative; otherwise the pipeline falls back to caption/context extraction (see `extract_problem_and_gates()` in `gate_problem_extractor.py`), where multi-character gates are matched directly and single-letter gates are accepted only with gate-indicating context. All detections are normalized to canonical names and returned as a unique sorted list (see the merge/normalization logic in `extract_problem_and_gates()`).

3.3.3 Identification of Quantum Algorithms or Problem Context

The quantum problem label associated with each circuit is identified using hierarchical keyword matching over the combined caption and nearby contextual text, rather than OCR, because algorithm names typically appear in prose (see the caption+context+title concatenation in `extract_problem_and_gates()` in `gate_problem_extractor.py`). The classifier applies a tiered decision logic that prioritizes specific named algorithms first (e.g., Grover's, Shor's, VQE, QAOA, Deutsch–Jozsa), then falls back to functional categories (e.g., Quantum Fourier Transform, arithmetic operations, quantum phase estimation), and finally to broader problem classes (e.g., magic state injection, oracle gate implementation, state preparation, circuit simulation) (see `classify_problem()` in `gate_problem_extractor.py`). If no reliable keyword match is found, the label is set to "Unknown" to avoid speculative annotation.

3.3.4 Extraction and Position Mapping of Descriptive Text

Descriptive text is extracted using a caption-first strategy. Captions are cleaned and filtered by removing figure prefixes and formatting artifacts, rejecting equation-like strings, and applying a relevance score to suppress generic/bibliographic text (see `clean_description_text()`, `is_math_like()`, and `calculate_semantic_relevance()` in `text_layout_utils.py`). If no usable caption is obtained, the pipeline falls back to page-local paragraphs within a fixed vertical window around the figure (see `extract_paragraphs_around_figure()` and `select_description_sentences()` in `text_layout_utils.py`).

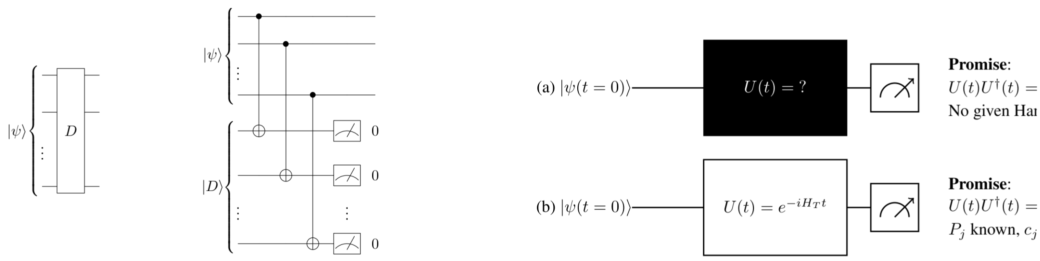
For traceability, each selected description is mapped to character offsets in the document-wide `full_text` produced by `build_full_text()` in `text_layout_utils.py`, and the best (begin,end) span is found via normalized matching with a fuzzy fallback (see `find_sentence_position_best()` in `build_dataset.py`), which is stored in `text_positions`.

4 Error Analysis and Failure Modes

This section summarizes the dominant failure modes observed during dataset compilation. Errors fall into three groups: (i) image-level misclassification (false positives/negatives), (ii) metadata extraction noise, and (iii) PDF parsing failures that prevent some papers from being processed.

4.1 Image-level Misclassification (False Positives/Negatives)

Most false positives come from figures that share the same low-level geometry as circuits: line-dense plots (gridlines/waveforms), box-and-arrow schematics (block diagrams), tables/matrix layouts, and other white-background graphics with repeated straight lines. A second frequent cause is mixed crops, where caption/prose is partially included in the extracted region, making the image look more "technical" and increasing the chance of being classified as a circuit. False negatives mainly occur when circuits are too small (tiny insets or subpanels), embedded inside composite figures together with other plots/diagrams, or drawn in a highly stylized/non-standard way that differs from common training examples. In addition, circuits can be missed when vector drawing content is fragmented across multiple clusters, leading to incomplete or noisy candidate crops that are later rejected.



(a) True positive (b) False positive

Figure 2: Qualitative error examples from the extraction pipeline.

4.2 Metadata Extraction Noise

Even with a correct circuit image, metadata can be imperfect:

- Unknown problem/algorithm labels arise when keywords are not stated near the figure or the image is a false positive, so surrounding text is not circuit-related.

- Description misalignment occurs because captions can be minimal and nearby paragraphs often discuss theory; page-local extraction is therefore an approximation.
- Gate noise mainly comes from OCR errors on small fonts and ambiguity of single-letter tokens that may be variables rather than gates.
- Figure-number ambiguity (e.g., “3a”, “(b)”, “S2”) requires deterministic integer normalization, which cannot perfectly preserve the original label.

Example: false positive leading to metadata noise.


```
"arXiv_2507.22874_page6_clusters1.png": {
    "arxiv_number": "arXiv:2507.22874",
    "page_number": 6,
    "figure_number": -1,
    "quantum_gates": [
      "X"
    ],
    "quantum_problem": "Unknown",
    "descriptions": [
      "tunneling amplitudes [37], to reach a phase with well-defined X2,3",
      "We emphasize that these atomic limit Dirac strings per- sist in the system
      "Con- sequently, a given phase of an Euler system with non- trivial BZBCs wi
    ],
    "text_positions": [
      [
        27240,
        27307
      ],
      [
        27963,
        28232
      ],
      [
        28232,
        28456
      ]
    ]
  },
```


This record illustrates how a false positive propagates into noisy metadata: the gate extractor returns `X`, but in this image `X` is a variable label rather than a quantum gate. The problem label remains `Unknown`, the figure number is -1 because no numeric caption identifier was detected, and the extracted descriptions are generic/non-informative with respect to quantum circuits.

4.3 PDF parsing failures

A small subset of PDFs failed mid-processing due to non-standard or malformed PDF content. Typical failures included color space parsing errors and missing graphics-state resources (e.g., missing ExtGState entries like pgf@ca1, gs0), which can occur in PDFs generated by complex TeX/PGF pipelines. These failures are unavoidable in large-scale arXiv mining and must be tolerated by skipping the affected papers and continuing the ordered processing.

4.4 Non-Adopted Techniques and Trade-offs

Several stricter rules improved apparent precision in small runs but were not adopted because they are hard-coded heuristics that do not generalize across diverse paper styles and are therefore not scalable for large-scale dataset construction:

- Discard if no gates are detected (OCR/text): reduces some false positives, but removes many valid circuits with tiny, symbolic, or unreadable labels, sharply increasing false negatives.
- Discard if gates exist but description and problem/algorithm are empty: sometimes yields "cleaner" outputs, but fails when explanations are not local to the figure, increasing false negatives and requiring many more papers to reach the target size.
- Pure heuristics-only classification (no CNN): required frequent manual threshold tuning and still misfired on circuit-like non-circuit figures (plots/flowcharts/tables).
- Aggressive cropping/splitting: occasionally isolates circuits but becomes unstable at scale, increases fragmentation, and reduces recall and metadata consistency.

5 Dataset Quality Assessment and Validation

Dataset quality is assessed along three dimensions: correctness, consistency, and traceability.

Correctness and Precision of Circuit Images

The final dataset contains 250 circuit images, obtained after processing approximately 470 PDFs in the prescribed order. To estimate correctness, I manually inspected a sample of the extracted images and found that approximately 60–65% of the saved images are clearly valid quantum circuit diagrams (precision estimate). The remaining errors are primarily false positives caused by figures that visually resemble circuits, such as line-dense plots, box-and-arrow schematics, tables with strong grid structure, or mixed crops where nearby text is partially included. False negatives (missed circuits) also exist, primarily for tiny inset circuits, multi-panel composite figures, and stylized circuit notation that does not match typical circuit structure.

Consistency and Schema Compliance

The dataset is designed to remain structurally consistent even when extraction is uncertain. All records follow the required JSON schema and keep field types stable. In particular:

- page_number is always an integer, using footer-based extraction with a fallback when the printed page number is missing.
- figure_number is always an integer, using deterministic normalization for labels like “3a/3.1”, and a sentinel integer for labels that cannot be represented as a numeric figure id.
- quantum_gates is normalized to canonical gate names and deduplicated, so the same gate is not listed in multiple variants.
- Descriptions are non-empty in most cases, because the pipeline uses a caption-first strategy with a fallback to nearby page-local paragraphs when captions are missing or unusable.
- text_positions are consistently stored as (begin, end) pairs, enabling machine-checkable traceability to the reconstructed document text.

Quality Controls and Observed Improvements

Several pipeline controls were introduced specifically to improve dataset quality:

- Text-block rejection reduced obvious false positives where clustered regions were actually prose, bibliography, or footers.
- Crop trimming reduced caption leakage and removed “Figure ... ” lines that were accidentally included in candidate regions.
- OCR-first gate extraction with strict acceptance rules reduced gate noise, especially for single-letter gates that otherwise collide with variables and labels.
- Caption-first description selection with math filtering improved description relevance by avoiding equation-dominated or purely referential text.
- Conservative “Unknown” labeling improved semantic consistency by avoiding speculative algorithm/problem assignment when keywords were missing or ambiguous.

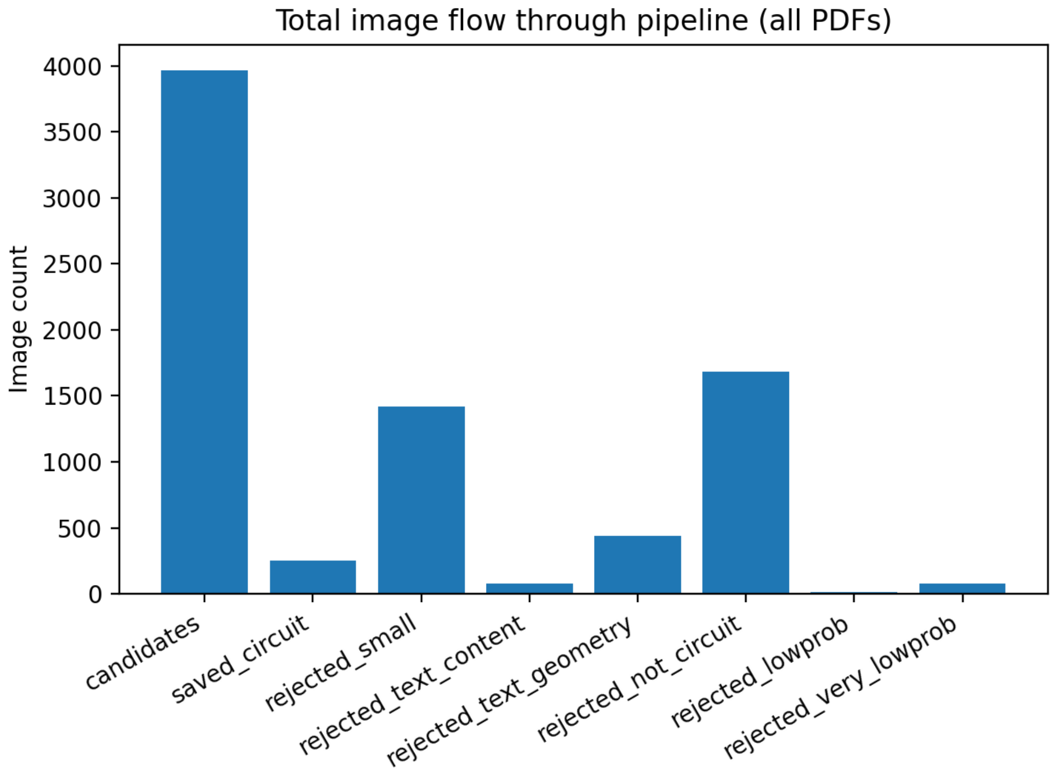


Figure 3: Bar graph of candidate filtering and final saved circuits.

Filtering Impact and Rejection Statistics

Figure 3 summarizes the end-to-end flow of candidate regions through the extraction pipeline across all processed PDFs. Starting from all candidate regions produced by clustering and merging vector drawing elements ("candidates"), the pipeline applies successive rejection gates: very small regions are removed first, then regions that behave like prose/bibliography are filtered using content-based and geometry-based text detectors, and finally the remaining ambiguous candidates are evaluated by the circuit vs non-circuit CNN. At the model stage, many candidates are rejected either because they are predicted as non-circuits or because their circuit probability falls below conservative confidence thresholds ("lowprob" and "very_lowprob"), and only the small subset that survives all rejection criteria is retained and written to disk as "saved_circuit". This distribution provides a quantitative sanity check that the dataset is produced through layered quality control rather than by saving all extracted graphics.

6 Dataset Feasibility and Limitations

Fully automated collection of a quantum-circuit image–text dataset from arXiv is feasible in principle because the pipeline processes papers deterministically from an ordered list, applies layered filtering to control noise, and outputs schema-consistent, traceable records without manual annotation. The main scaling bottlenecks are (i) heterogeneous PDF layouts and author toolchains that cause occasional parsing/rendering failures, (ii) persistent visual ambiguity between circuits and other structured scientific figures that limits achievable precision without stronger models or richer layout understanding, and (iii) metadata locality issues where algorithms and descriptions are not stated near the figure, leading to "Unknown" or weak descriptions. In practice, the approach can scale, but quality will plateau unless additional robustness measures are added.

Limitations

Even after applying quality controls, some limitations remain: a subset of circuit-like scientific figures (plots, block diagrams, tables) still slip through as false positives and some genuine circuits are missed due to tiny size, composite layouts, or unusual styling; metadata can be noisy because OCR occasionally misreads small gate labels and algorithm/description text is not always local to the figure; and a small number of papers fail mid-processing due to non-standard or malformed PDF structures that trigger rendering/resource errors, which is an unavoidable issue when mining large, heterogeneous arXiv corpora.

7 Future Work

Future improvements should focus on increasing robustness across the diversity of arXiv PDFs and reducing circuit/non-circuit ambiguity. On the vision side, training the circuit classifier on a larger and more diverse set of circuit styles and hard negative examples (plots, block diagrams, tables) would improve generalization and reduce false positives, and a lightweight detector/segmenter could help isolate circuits inside composite figures. On the document side, stronger figure–caption association and better region segmentation would reduce caption leakage and improve description relevance, while expanding gate extraction with higher-quality OCR preprocessing and additional gate patterns could reduce label noise. Finally, adding systematic PDF error handling (skip/repair strategies and fallback rendering) would increase scalability by preventing malformed PDFs from interrupting large-scale runs.

8 Conclusion

This project demonstrates that it is feasible to compile a structured image–text dataset of quantum circuit diagrams from arXiv quant-ph publications using a fully automated and reproducible pipeline. By combining layout-driven candidate extraction, layered filtering, CNN-based circuit classification, and caption/context-based metadata extraction, the pipeline produces schema-consistent records with traceable links back to source pages and text spans, and it successfully generated a dataset of 250 circuit images from roughly 470 papers. At the same time, the results highlight clear limits of automation in heterogeneous scientific PDFs: circuit-like non-circuit figures still cause false positives, small or composite circuits can be missed, OCR introduces occasional gate noise, and algorithm/description fields are sometimes weak because relevant explanations are not local to the figure. Overall, the approach scales in principle, but maintaining high precision and rich metadata at larger scale will require stronger figure segmentation, more diverse classifier training, and improved robustness to PDF parsing variability.

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